

Will Epperson

willepperson.com willepp@live.com github.com/willeppy
linkedin.com/in/willepperson Google Scholar Seattle, WA

I am a research engineer on the AI Frontiers team at Microsoft Research. I work on personal assistant agents, in particular always-on agents that collaborate in multi-agent systems. I build the harnesses and communication platforms that agents run on, and the evaluations, environments, and debugging tools that measure whether an agent acts in its user's interest.

RECENT PROJECTS

Agora: Network of Personal Assistant Agents

Microsoft Research, 2026
[Blog](#)

Built Agora, a platform where personal assistant agents collaborate on behalf of their users. Technical lead on the agent communication platform, the agent harness, and the evals that measure how well agents perform. Built an RL evaluation environment for multi-agent calendar scheduling. This enabled researchers to run experiments on a live network of 100+ always-on agents (see [blog](#)).

SocialReasoning-Bench

Microsoft Research, 2026
[Blog](#) · [Code](#)

Co-designed and built a benchmark that measures whether an agent acts in its user's best interest when negotiating on their behalf, in calendar coordination and marketplace settings. Worked on the experiment infrastructure, data generation pipelines, task design, and metric design.

Magentic Marketplace

Microsoft Research, 2025
[Paper](#) · [Code](#)

Built the experiment infrastructure for an open-source simulation environment for two-sided agentic markets, where assistant agents buy from competing business agents.

AGDebugger

CHI 2025
[Paper](#) · [Code](#)

Designed, implemented, and evaluated an interactive debugging system for multi-agent systems that lets developers inspect agent traces and reset agents back to previous state.

EXPERIENCE

Microsoft Research · Senior Software Engineer, AI Frontiers

Aug 2025 to present

- Research engineer working on AI agents, multi-agent systems, evaluations, and environments (see [Recent Projects](#)).
- Mentored two PhD student interns on projects around CUA benchmarks and multi-agent theory of mind.

Microsoft Research · Research Intern, AI Frontiers

Summer 2024

- Built AGDebugger, an interactive debugging tool for multi-agent AI systems ([CHI 2025](#)).

Databricks · Software Engineering Contractor

Summer 2022

- Designed and shipped a production feature for creating dashboards by specifying fields of interest in a dataset ([Best Paper at VDS 2023](#)).

Microsoft Research · Research Intern, VIDA group

Summer 2021

- Led a study of reuse and sharing among data scientists in software teams ([ICSE 2022](#)). Co-authored Diff in the Loop, a tool for comparing data frames during analysis ([CHI 2022](#)).

Point72 Asset Management · Data Analytics Intern, Market Intelligence

Summer 2019

- Cleaned, modeled, and analyzed large alternative datasets.

EDUCATION

Ph.D. in Human-Computer Interaction · Carnegie Mellon University, Pittsburgh, PA

2020 to 2025

- Thesis: [Interactive Data Profiling](#). Advisors: Dominik Moritz and Adam Perer.
- My PhD focused on interactive data profiling systems that help data scientists and AI developers better understand their data, leading to open-source tools adopted by the Python community and publications and awards at top computer science venues. Mentored five undergraduate and masters students on research projects.

B.S. in Computer Science · Georgia Institute of Technology, Atlanta, GA

2016 to 2020

- GPA 4.0, Summa Cum Laude. Threads in Intelligence and Modeling/Simulation. Stamps President's Scholar (full scholarship, 40 per class).
- Undergraduate researcher in the Polo Club of Data Science. Built FairVis, visual analytics for discovering intersectional bias in ML models ([IEEE VAST 2019](#)).

PUBLICATIONS

SocialReasoning-Bench: Measuring Whether AI Agents Act in Users' Best Interests

Tyler Payne, **Will Epperson**, Safoora Yousefi, Zachary Huang, Gagan Bansal, Wenyue Hua, et al. Microsoft Research Blog, 2026. [Blog](#). [Code](#).

Whimsical Strategies Break AI Agents: Generating Out-of-Distribution Adversarial Strategies at Scale

Zachary Huang, Tyler Payne, Gagan Bansal, **Will Epperson**, Wenyue Hua, Adam Fourney, et al. Microsoft Research Blog, 2026. [Blog](#).

Red-Teaming a Network of Agents: Understanding What Breaks When AI Agents Interact at Scale

Gagan Bansal, Shujaat Mirza, Keegan Hines, **Will Epperson**, Zachary Huang, Whitney Maxwell, et al. Microsoft Research Blog, 2026. [Blog](#).

Magentic Marketplace: An Open-Source Environment for Studying Agentic Markets

Gagan Bansal, Wenyue Hua, Zezhou Huang, Adam Fourney, Amanda Swearngin, **Will Epperson**, et al. 2025. [PDF](#). [Code](#).

Interactive Data Profiling

Will Epperson. PhD thesis, Carnegie Mellon University, 2025. [PDF](#).

Texture: Structured Exploration of Text Datasets

Will Epperson, Arpit Mathur, Dominik Moritz, Adam Perer. 2025. [PDF](#). [Code](#).

Interactive Debugging and Steering of Multi-Agent AI Systems

Will Epperson, Gagan Bansal, Victor Dibia, Adam Fourney, Jack Gerrits, Erkang Zhu, Saleema Amershi. CHI Conference on Human Factors in Computing Systems, 2025. PDF. Code.

Over-Relying on Reliance: Towards Realistic Evaluations of AI-Based Clinical Decision Support

Venkatesh Sivaraman, Katelyn Morrison, **Will Epperson**, Adam Perer. CHI Workshop on HCI+Health, 2025. PDF.

Guided Statistical Workflows with Interactive Explanations and Assumption Checking

Yuqi Zhang, Adam Perer, **Will Epperson**. VIS 24: IEEE Conference on Data Visualization, 2024. PDF. Code.

Dead or Alive: Continuous Data Profiling for Interactive Data Science **Best Paper Honorable Mention**

Will Epperson, Vaishnavi Gorantla, Dominik Moritz, Adam Perer. VIS 23: IEEE Conference on Data Visualization, 2023. PDF. Code.

A Declarative Specification for Authoring Metrics Dashboards **Best Paper**

Will Epperson, Kanit Wongsuphasawat, Allison Whilden, Fan Du, Justin Talbot. VDS at VIS 23: Visual Data Science Symposium, 2023. PDF.

Leveraging Analysis History for Improved In Situ Visualization Recommendation

Will Epperson, Doris Jung-Lin Lee, Leijie Wang, Kunal Agarwal, Aditya Parameswaran, Dominik Moritz, Adam Perer. EuroVis 22: Eurographics Conference on Visualization, 2022. PDF. Code.

Strategies for Reuse and Sharing among Data Scientists in Software Teams

Will Epperson, April Yi Wang, Robert DeLine, Steven M. Drucker. ICSE 22: ACM International Conference on Software Engineering, 2022. PDF.

Diff in the Loop: Supporting Data Comparison in Exploratory Data Analysis

April Yi Wang, **Will Epperson**, Robert DeLine, Steven M. Drucker. CHI Conference on Human Factors in Computing Systems, 2022. PDF.

RECAST: Interactive Auditing of Automatic Toxicity Detection Models

Austin P. Wright, Omar Shaikh, Haekyu Park, **Will Epperson**, Muhammed Ahmed, Stephane Pinel, et al. 24th ACM Conference on Computer-Supported Cooperative Work & Social Computing, 2021. PDF.

FairVis: Visual Analytics for Discovering Intersectional Bias in Machine Learning

Angel Cabrera, **Will Epperson**, Fred Hohman, Minsuk Kahng, Jamie Morgenstern, Duen Horng (Polo) Chau. IEEE Conference on Visual Analytics Science and Technology, 2019. PDF. Blog. Code.

Full list at willepperson.com/cv.

HONORS, SERVICE, SKILLS

Awards Best Paper Honorable Mention, IEEE VIS 2023. Best Paper, VDS 2023. President's Undergraduate Research Award, Georgia Tech 2019.

Reviewing CHI 2022, 2023, 2025. IEEE VIS 2021 to 2024. CSCW 2021.

Teaching TA for Programmable User Interfaces and Interactive Data Science (CMU), Intro to Database Systems (Georgia Tech).

Skills Python, TypeScript/JavaScript, SQL. PyTorch, scikit-learn, Svelte, React, D3, Vega-Lite. English (native), Spanish (advanced).